

Memory Management MCQs

1. What type of memory is volatile and used for storing currently executing processes?

- A) Secondary Memory
- B) Cache Memory
- C) Primary Memory (RAM)
- D) Virtual Memory

Answer: C) Primary Memory (RAM)

2. Which of the following is an example of secondary memory?

- A) DDR4 RAM
- B) L1 Cache
- C) Hard Drive (HDD)
- D) CPU Registers

Answer: C) Hard Drive (HDD)

3. What is the main advantage of cache memory?

- A) It is non-volatile.
- B) It is located far from the CPU.
- C) It stores frequently accessed data.
- D) It has unlimited size.

Answer: C) It stores frequently accessed data.

4. What does virtual memory allow?

- A) Execution of programs larger than physical memory.
- B) Faster access to data.
- C) Permanent storage of data.
- D) Direct access to the CPU.

Answer: A) Execution of programs larger than physical memory.

5. Which memory allocation strategy allocates the first available block of memory that is large enough?

- A) Best Fit
- B) Worst Fit
- C) First Fit
- D) Dynamic Allocation

Answer: C) First Fit

6. What is a disadvantage of the Best Fit memory allocation strategy?

- A) It is too fast.
- B) It leads to external fragmentation.
- C) It is slower than other strategies.
- D) It wastes large memory chunks.

Answer: C) It is slower than other strategies.

7. What is internal fragmentation?

- A) Free memory exists but is not contiguous.
- B) Unused memory within allocated blocks.
- C) Memory allocated to a process is larger than needed.

- D) Memory is divided into fixed-sized partitions.

Answer: B) Unused memory within allocated blocks.

8. Which of the following is a characteristic of segmentation?

- A) Divides memory into fixed-sized blocks.
- B) Each segment is stored in a different memory location.
- C) Eliminates external fragmentation.
- D) Uses a single memory block for all processes.

Answer: B) Each segment is stored in a different memory location.

9. What does a page table do in paging?

- A) Maps logical page numbers to physical frame numbers.
- B) Stores the size of a segment.
- C) Combines free memory blocks.
- D) Allocates memory dynamically.

Answer: A) Maps logical page numbers to physical frame numbers.

10. What is the purpose of a dirty bit?

- A) To indicate if a page has been accessed.
- B) To indicate if a page has been modified in memory but not written back to disk.
- C) To track the size of memory blocks.
- D) To manage device drivers.

Answer: B) To indicate if a page has been modified in memory but not written back to disk.

11. What is the main function of I/O management?

- A) To allocate memory to processes.
- B) To coordinate the flow of data between the CPU, memory, and peripherals.
- C) To manage cache memory.
- D) To handle fragmentation.

Answer: B) To coordinate the flow of data between the CPU, memory, and peripherals.

12. Which of the following strategies may create larger remaining blocks?

- A) First Fit
- B) Best Fit
- C) Worst Fit
- D) Dynamic Allocation

Answer: C) Worst Fit

13. What is the main goal of memory compaction?

- A) To allocate memory dynamically.
- B) To reduce fragmentation by combining free memory blocks.
- C) To increase the size of cache memory.
- D) To manage I/O operations.

Answer: B) To reduce fragmentation by combining free memory blocks.

14. What is reentrant code?

- A) Code that cannot be shared among processes.
- B) Code that can be shared safely among multiple processes.

- C) Code that requires exclusive access to memory.
- D) Code that is executed only once.

Answer: B) Code that can be shared safely among multiple processes.

15. What is the primary disadvantage of continuous memory allocation?

- A) It is complex to implement.
- B) It suffers from internal fragmentation.
- C) It is too slow.
- D) It requires more memory than dynamic allocation.

Answer: B) It suffers from internal fragmentation.